DA10 Cleaning without hiding evidence

Week 4, Clean and chart | Session 10 of 24 | Python for Data Analysis

Today's question. What is wrong with the raw Sunrise sales file, and how many rows does each problem touch?

Objective. By the end: you can find missing values and duplicates, remove the duplicates on a copy, and save a cleaning log that says what you did.

> WHERE THIS IS USED
> Exports from tills often hold duplicates, blanks and typing slips. Good analysts keep a log of every fix so the owner can trust the result. How many problems can you find before you fix a single one?

Before this session: DA09 Sort, group, mean and median. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
raw data	The file exactly as it arrived. You never overwrite it.	The original receipt book.
missing value	A cell with nothing recorded. It is not the same as zero.	A blank in the register.
duplicate	A row that repeats another row exactly.	The same sale written twice.
data type	How a column is stored: whole number, decimal or text.	Whether a box holds numbers or words.
copy	A separate table to work on, so the raw table stays untouched.	A photocopy of the receipt book.
cleaning log	A record of each problem found and what was done about it.	The tailor's note of every change to a uniform.

Block A. Missing values and duplicates in a small table

Keep the raw table unchanged and work on a copy. isna marks missing cells True. Adding the True values counts them. drop_duplicates removes rows that repeat another row exactly.

Predict. Predict how many missing prices there are and how many rows stay after the duplicate is removed. Write your guess on paper first.

Your guess, written before you run it

```
import pandas as pd
raw = pd.DataFrame({"id": [1, 2, 2, 3], "price": [1000, None, None, 3000]})
clean = raw.drop_duplicates().copy()
```

```
print(raw["price"].isna().sum())
print(len(clean))
print(clean["price"].mean())
```

Run it, then compare. The raw table has 2 missing prices. After the duplicate row is removed, 3 rows stay and one price is still missing. The mean of the two known prices is 2000. The missing price is unknown and is not free.

Investigate.

1. Why would replacing the missing price with 0 change the meaning?
2. Could two identical rows be real separate sales?

Modify. Count how many rows in clean still have a missing price.

Block B. The raw Sunrise file

sunrise_sales_raw.csv is the Sunrise sales as an untidy export. Count the problems. to_csv saves a table to a file. index=False stops pandas adding an extra column of row numbers.

Predict. Predict how many duplicate rows the file holds. The clean version has 1513 sales. Write your guess on paper first.

Your guess, written before you run it

```
raw_sales = pd.read_csv("sunrise_sales_raw.csv")
print(raw_sales.shape)
print(raw_sales.duplicated().sum())
deduped = raw_sales.drop_duplicates()
print(deduped.shape)
print(deduped["units"].isna().sum())
log = pd.DataFrame({"problem": ["duplicate rows", "blank units"], "rows":
    [raw_sales.duplicated().sum(), deduped["units"].isna().sum()]})
log.to_csv("cleaning_log.csv", index=False)
print(pd.read_csv("cleaning_log.csv"))
```

Run it, then compare. The raw file has 1538 rows. 25 are exact duplicates, so removing them leaves 1513, the same as the clean file. 40 rows have blank units. The log table is saved and read back with two lines: 25 duplicate rows and 40 blank-unit rows.

Investigate.

1. Why does the cleaned row count of 1513 match the clean file?
2. Why not fill the 40 blank units with the mean?

Modify. Count the rows whose date is written with slashes, and add them to the log.

Make

Write a cleaning log for the four-row table in Block A: four raw rows, one duplicate removed, one price still missing. Save it as a CSV with to_csv. Write one sentence on why replacing the missing price with zero would change what the data says.

Save your work as DA10. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
SettingWithCopyWarning	You changed a slice of another table, so pandas is not sure which table you meant.	Make a real copy with .copy() before you change a table.
Row counts differ between steps	A step removed or added rows without you noticing.	Print len() after every step and write the numbers in the log.

Exit ticket

I keep the raw file untouched, work on a copy, and write down every change I make.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.